

Regime-Switching Reinforcement Learning for Portfolio Optimization

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Abstract

Financial markets exhibit strong non-stationarity and volatility clustering, which makes portfolio optimization difficult for traditional models that assume stable return distributions.

In this project, we study a regime-switching reinforcement learning framework for portfolio allocation. The approach combines a Hidden Markov Model (HMM) for market regime detection with a reinforcement learning agent that learns portfolio allocation policies from historical data.

At this milestone stage, we have implemented the core components of the system and integrated regime probabilities into the reinforcement learning environment. Preliminary experiments compare a baseline PPO portfolio agent with a regime-aware variant. This report summarizes the system design, implementation progress, and early experimental results.

1 Introduction

Portfolio optimization aims to allocate capital across a set of assets in order to balance expected return and risk. While classical approaches such as mean-variance optimization are well-studied, they rely on assumptions of stationarity and stable return distributions. In practice, financial markets exhibit strong non-stationarity, regime shifts, and volatility clustering, making static optimization approaches unreliable during turbulent periods.

Recent advances in reinforcement learning have enabled data-driven portfolio allocation strategies that adapt to observed market dynamics. However, most RL-based methods still train a single agent to operate across all market conditions, which can lead to poor performance during crisis periods.

2 Problem Statement and Motivation

The core problem addressed in this project is portfolio optimization under non-stationary and volatile market conditions. Specifically, we study whether explicitly incorporating market regimes into reinforcement learning-based portfolio optimization can improve robustness and stability compared to classical methods and single-agent RL approaches.

3 Related Work

Several prior works have explored reinforcement learning and planning-based approaches for portfolio optimization and financial decision-making.

Jiang et al. [1] propose a deep reinforcement learning framework for portfolio management using policy gradient methods, demonstrating improved performance over traditional benchmarks but assuming a single policy across all market conditions.

Moody and Saffell [4] present one of the earliest applications of reinforcement learning to trading systems, using recurrent reinforcement learning to directly optimize trading performance, though without explicit regime modeling.

Deng et al. [5] apply deep neural networks to financial trading, showing that representation learning can improve decision-making in complex markets, but still relying on a unified model across regimes.

Yang et al. [3] study reinforcement learning for portfolio allocation with risk-sensitive objectives, highlighting the importance of risk-aware reward design but not addressing regime shifts explicitly.

The FinPlan framework [2] introduces planning-based methods for financial decision-making, focusing on structured reasoning and long-term planning under uncertainty. While effective, these approaches do not explicitly train separate policies for different market regimes.

4 Limitations of Existing Approaches

Existing portfolio optimization methods suffer from several limitations:

- Strong reliance on stationarity assumptions.
- Single-policy RL models that struggle during high-volatility regimes.
- Limited analysis of drawdowns and stability across market conditions.

5 Proposed Approach

We propose a regime-switching reinforcement learning framework that combines market regime detection with reinforcement learning-based portfolio allocation. The objective is to incorporate information about market volatility regimes into the decision-making process of the reinforcement learning agent.

6 System Architecture

The overall system combines a market regime detection module with a reinforcement learning portfolio allocation agent. The goal is to allow the agent to adapt its strategy depending on market volatility conditions.

The architecture consists of four main stages:

1. **Data Preparation:** Historical ETF price data is collected and converted into daily log returns and rolling volatility features.
2. **Regime Detection:** A Gaussian Hidden Markov Model (HMM) is trained to classify market conditions into different volatility regimes.
3. **State Construction:** Regime probabilities inferred by the HMM are incorporated into the reinforcement learning state representation.

4. **Portfolio Allocation:** A reinforcement learning agent (PPO) outputs portfolio weights across the selected assets.

Pipeline summary:

Market Data $\rightarrow$ Feature Engineering $\rightarrow$ HMM Regime Detection $\rightarrow$ RL Portfolio Agent $\rightarrow$
Portfolio Allocation

7 Market Regime Detection

To capture changing market conditions, we model market regimes using a Gaussian Hidden Markov Model (HMM). The HMM learns latent market states from historical market statistics and outputs the probability of the market being in each regime at a given time step.

The model is trained using the Expectation-Maximization (EM) algorithm. Input features used for regime detection include:

- Daily log returns
- Rolling volatility of returns
- Moving average ratios

The model identifies three regimes corresponding to different volatility environments:

- Low volatility regime
- Medium volatility regime
- High volatility regime

Model selection experiments were conducted using the Bayesian Information Criterion (BIC), which indicated that three regimes provide the best balance between model complexity and explanatory power.

The output of the HMM is a vector of regime probabilities:

$$[p_{low}, p_{medium}, p_{high}]$$

These probabilities are incorporated as additional features in the reinforcement learning environment.

8 Reinforcement Learning Environment

The portfolio allocation problem is formulated as a reinforcement learning task. A custom trading environment was implemented using the Gymnasium framework.

At each time step the agent observes a state vector consisting of:

- Recent asset returns
- Rolling volatility features
- Previous portfolio weights

- Regime probabilities inferred by the HMM

The action space represents portfolio allocation weights across the selected assets. The reward function is defined as:

$$R_t = w_t \cdot r_t - \lambda |w_t - w_{t-1}|$$

where:

- w_t represents portfolio weights
- r_t represents asset returns
- λ penalizes portfolio turnover

This reward formulation encourages the agent to maximize portfolio return while controlling excessive trading.

9 Experimental Setup

Experiments were conducted using a portfolio of major exchange traded funds (ETFs):

- SPY (S&P 500)
- QQQ (NASDAQ)
- IWM (Russell 2000)
- TLT (Treasury Bonds)
- GLD (Gold)

Historical data was split into training and testing periods:

- Training period: 2008 – 2018
- Testing period: 2019 – 2024

Two reinforcement learning models were trained:

- Baseline PPO agent (without regime information)
- Regime-aware PPO agent (with HMM regime probabilities)

Both models were implemented using the Stable-Baselines3 PPO framework.

10 Preliminary Results

Initial experiments compare the performance of the baseline PPO agent and the regime-aware PPO agent.

The regime-aware model achieves higher cumulative return and improved risk-adjusted performance. These preliminary results suggest that incorporating regime information may help the reinforcement learning agent adapt more effectively to changing market conditions.

Model	Total Return	Sharpe Ratio	Max Drawdown
Baseline PPO	0.92	0.81	-0.289
Regime-Aware PPO	1.24	0.96	-0.272

Table 1: Preliminary comparison between baseline and regime-aware PPO agents

11 Future Work

This milestone focuses on building and integrating the regime-aware reinforcement learning framework. Several extensions remain for future work:

- Train regime-specialized reinforcement learning agents for different volatility regimes.
- Explore alternative reward functions such as Sortino ratio and CVaR-based objectives.
- Compare additional reinforcement learning algorithms including SAC and TD3.
- Incorporate additional market signals such as macroeconomic indicators and volatility indices.
- Conduct robustness testing across different datasets and market conditions.

12 Conclusion

This milestone report presents the design and implementation of a regime-aware reinforcement learning system for portfolio optimization. The system integrates Hidden Markov Model based regime detection with a PPO-based reinforcement learning portfolio allocation agent.

Preliminary experiments indicate that incorporating regime information may improve risk-adjusted performance compared to a standard reinforcement learning approach. Future work will focus on expanding the experimental analysis and exploring more advanced regime-aware learning architectures.

A Individual Contributions

Archit Shukla

- Designed and implemented the reinforcement learning portfolio environment.
- Implemented the PPO training pipeline using Stable-Baselines3.
- Integrated regime probabilities into the RL state representation.
- Conducted baseline and regime-aware training experiments.
- Evaluated model performance and generated experimental results.

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- Implemented the Hidden Markov Model for market regime detection.
- Designed the volatility-based regime classification methodology.
- Conducted experiments for regime selection and evaluation.
- Developed baseline comparisons with classical portfolio optimization approaches.

References

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